

Transfer Learning

THE ABILITY OF A SYSTEM TO RECOGNIZE AND APPLY KNOWLEDGE
AND SKILLS LEARNED IN PREVIOUS TASKS TO NOVEL TASKS

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Transfer Learning

- What is Transfer Learning?
- When to Use Transfer Learning?
- How to Use Transfer Learning?
- Advantages and Limitations
- What is the difference between Transfer Learning and parameters tuning?

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Transfer Learning

Learning to learn, life-long learning, knowledge transfer, multitask learning, knowledge consolidation, context-sensitive learning, knowledge-based inductive bias, metalearning, and incremental/cumulative learning

Transfer learning:

- In 2005,
- By the Broad Agency Announcement (BAA) 05-29 of Defense Advanced Research Projects Agency (DARPA)'s Information Processing Technology Office (IPTO)
- “the ability of a system to recognize and apply knowledge and skills learned in previous tasks to novel tasks”.
- With this definition, transfer learning aims to extract the knowledge from one or more source tasks and applies the knowledge to a target task.
- In contrast to multitask learning, rather than learning all of the source and target tasks simultaneously, transfer learning cares most about the target task. The roles of the source and target tasks are no longer symmetric in transfer learning.

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Transfer learning

- the knowledge of an already trained machine learning model is applied to a different but related problem. For example, if you trained a simple classifier to predict whether an image contains a backpack, you could use the knowledge that the model gained during its training to recognize other objects like sunglasses.
- try to exploit what has been learned in one task to improve generalization in another. We transfer the weights that a network has learned at “task A” to a new “task B.”
- is mostly used in computer vision and natural language processing tasks like sentiment analysis due to the huge amount of computational power required.
- is not a ML technique, but can be seen as a “design methodology”
- it has become popular in combination with NN require huge amounts of data and computational power.

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Transfer learning

- is a technique in ML where a model trained on one task is re-purposed on a second related task.
- is an optimization that allows rapid progress or improved performance when modeling the second task.
- is the improvement of learning in a new task through the transfer of knowledge from a related task that has already been learned.
- is related to problems such as multi-task learning and concept drift and is not exclusively an area of study for deep learning.
- is popular in DL given the enormous resources required to train deep learning models or the large and challenging datasets on which deep learning models are trained.
- only works in deep learning if the model features learned from the first task are general.

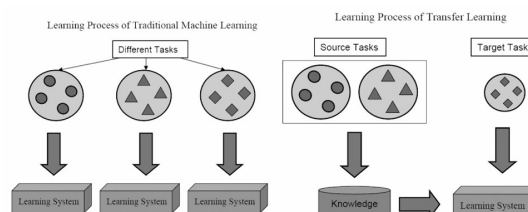
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Transfer Learning

Transfer learning, used in machine learning, is the reuse of a pre-trained model on a new problem.

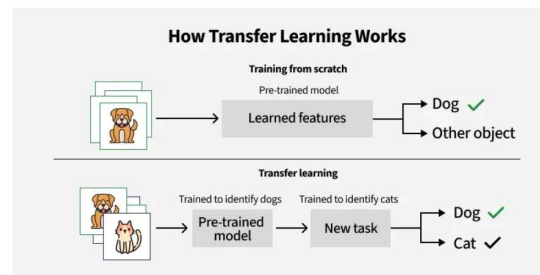
- a machine exploits the knowledge gained from a previous task to improve generalization about another.
- For example, in training a classifier to predict whether an image contains food, you could use the knowledge it gained during training to recognize drinks.
- Transfer learning **is the reuse of a pre-trained model on a new problem**. It's popular in deep learning because it can train deep neural networks with comparatively little data. This is very useful in the data science field since most real-world problems typically do not have millions of labeled data points to train such complex models.



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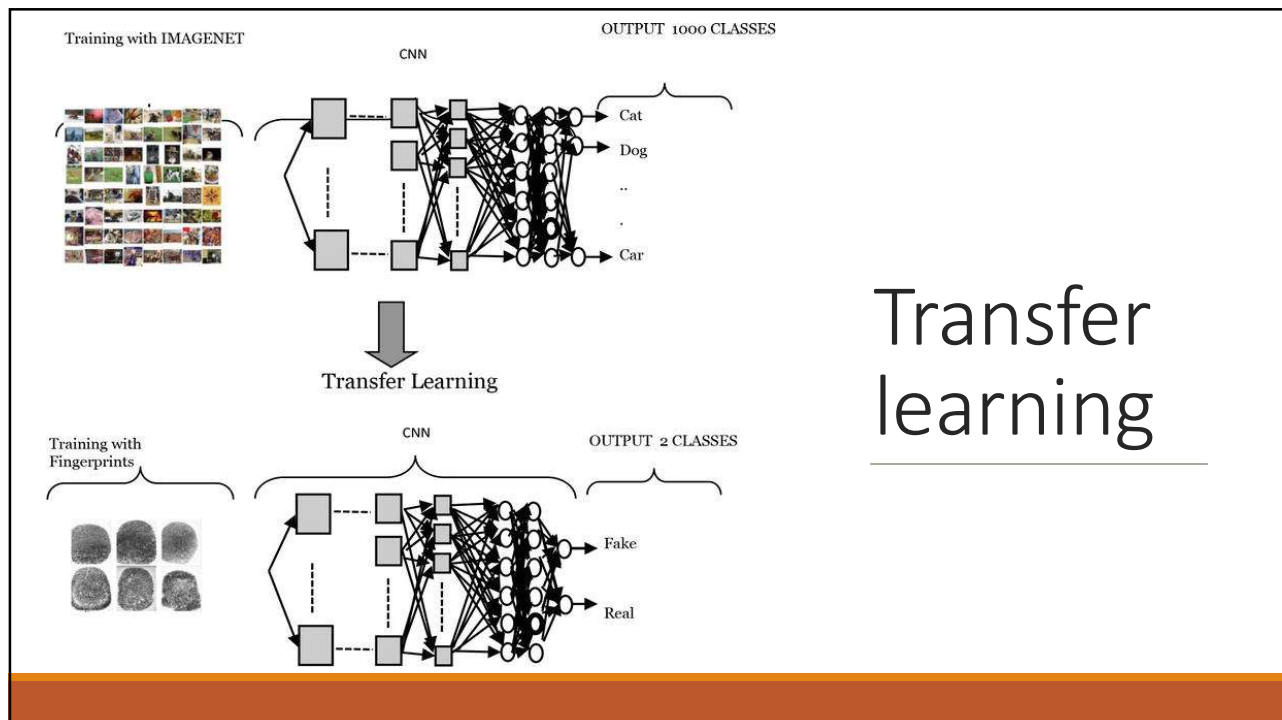
Why Use Transfer Learning ?

- saving training time
- better performance of neural networks (in most cases)
- not needing a lot of data
- one of the most promising techniques that could lead to artificial general intelligence (AGI)



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When to Use Transfer Learning?

- **Lack of training data:** There isn't enough labeled training data to train your network from scratch.
- **Existing network:** There already exists a network that is pre-trained on a similar task, which is usually trained on massive amounts of data.
- **Same input:** When task 1 and task 2 have the same input.

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Why Use Transfer Learning ?

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Approaches to Transfer Learning

- Training a model to reuse it
- Use an already pre-trained model
- Feature extraction

Some popular **pre-trained** models are already available

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Types of Transfer Learning

There are three adjacent practices or sub-settings of transfer learning. The difference between three adjacent practices result from changes in the relationship between the source domain, target domain, and tasks to be completed:

- **Inductive transfer** - when the source and target tasks are different, regardless of any difference or similitude between the target and source domains (i.e. datasets). This can manifest in computer vision models when architectures pretrained for feature extraction on large datasets are then adopted for further training on a specific task, such as object detection. Multitask learning, which consists of simultaneously learning two different tasks (such as image classification and object detection) on the same dataset, can be considered a form of inductive transfer.

- **Unsupervised learning** - similar to inductive transfer, as the target and source tasks are different. But in inductive transfer, source and/or target data is often labeled. Per its name, unsupervised transfer learning is unsupervised, meaning there is no manually labeled data. Inductive transfer can be considered supervised learning. One common application of unsupervised learning is fraud detection. By identify common patterns across an unlabeled dataset of transactions, a model can further learn to identify deviating behaviors as possible fraud.

- **Transductive transfer** - when the source and target tasks are the same, but the datasets (or domains) are different. The source data is typically labelled while the target data is unlabeled. Domain adaptation is a form of transductive learning, as it applies knowledge gained from performing a task on one data distribution towards the same task on another data distribution. An example of transductive transfer learning is the application of a text classification model trained and tested on restaurant reviews to classify movie reviews.

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Transfer learning vs finetuning

Transfer learning and **finetuning** reuse preexisting machine learning models as opposed to training new models.

Transfer learning is distinct from finetuning:

- **Finetuning** refers to the process of further training a model on a task-specific dataset to improve performance on the initial, specific task for which the model was built.
- **Transfer learning** signifies when users adapt a model to a new related problem as opposed to the same problem.

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Transfer Learning

Transfer learning is a machine learning technique where a model trained on one task or dataset is reused as a starting point for a different but related task.

Instead of starting the learning process from scratch, transfer learning leverages the knowledge gained from solving one problem and applies it to a new, possibly unrelated problem.

This approach is particularly useful when the new task has a smaller dataset or when computational resources are limited.

In transfer learning, the pre-trained model is typically trained on a large dataset for a related task, such as image classification on ImageNet or natural language processing on a large text corpus.

The knowledge captured by the source model, in the form of learned features and representations, is then transferred to the new model, known as the "target model," which is trained on the target task with a smaller dataset.

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Faster Training

By leveraging pre-trained models, transfer learning can significantly reduce the time and computational resources required to train a new model from scratch, especially when dealing with large datasets.



Improved Performance

Transfer learning allows the new model to benefit from the knowledge captured by the pre-trained model on a related task, leading to better generalization and performance, particularly when the target dataset is small.



Addressing Data Scarcity

In many real-world scenarios, obtaining labeled data for training machine learning models can be challenging and expensive. Transfer learning enables the reuse of knowledge learned from larger datasets, mitigating the effects of data scarcity.

Benefits Transfer learning

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Advantages

- **Computational costs** - reduces the requisite computational costs to build models for new problems. By repurposing pretrained models or pretrained networks to tackle a different task, users can reduce the amount of model training time, training data, processor units, and other computational resources.
- **Dataset size** - helps alleviate difficulties involved in acquiring large datasets. For instance, large language models (LLMs) require large amounts of training data to obtain optimal performance. Quality publicly available datasets can be limited, and producing sufficient manually labelled data can be time-consuming and expensive.
- **Generalizability** - . While transfer learning aids model optimization, it can further increase a model's generalizability. Because transfer learning involves retraining an existing model with a new dataset, the retrained model will consist of knowledge gained from multiple datasets. It will potentially display better performance on a wider variety of data than the initial base model trained on only one type of dataset.

Preprocessing techniques and feature engineering, such as data augmentation and feature extraction, are still necessary when using transfer learning.

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Disadvantages

Transfer learning works better when three conditions are met:

- both learning tasks are similar
- source and target datasets data distributions do not vary too greatly
- a comparable model can be applied to both tasks

When these conditions are not met, transfer learning can negatively affect model performance. Literature refers to this as **negative transfer**.